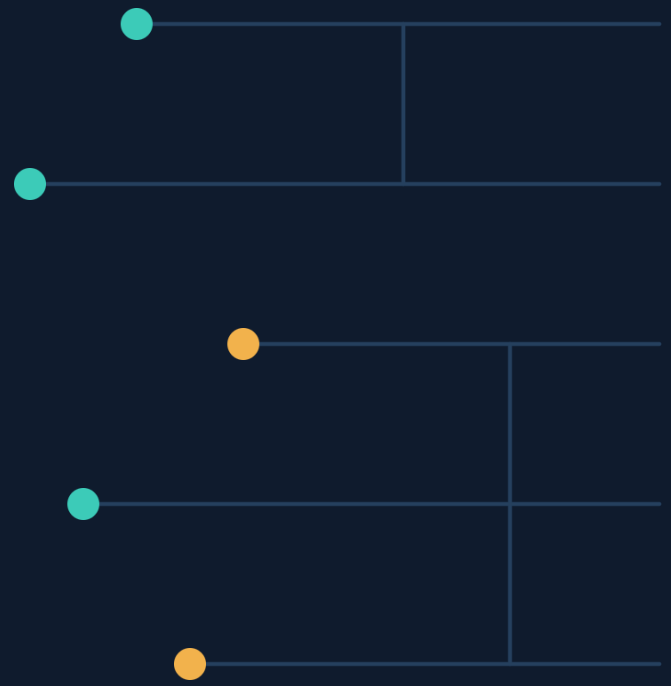
VERILOG HDL · RV32I

RISC-V RV32I Single-Cycle Processor



Abhishek(2024BITE060)
Bharat Bushan (2024BITE054)

OVERVIEW

A Complete Single-Cycle RV32I CPU

Every instruction is fetched, decoded, executed, written to memory or registers, and the next PC is chosen, all within one clock cycle.

37
RV32I instructions implemented

12
Verilog modules including the top level

32
32-bit registers, x0 fixed at zero

1
Clock cycle per instruction

INSTRUCTION MEMORY

256 words, word-aligned, preloaded with NOPs

DATA MEMORY

1024 bytes, byte-addressable, little-endian

TOP-LEVEL PORTS

Inputs clk and reset; outputs pc_out and instruction_out

INSTRUCTION SET

Supported RV32I Instructions

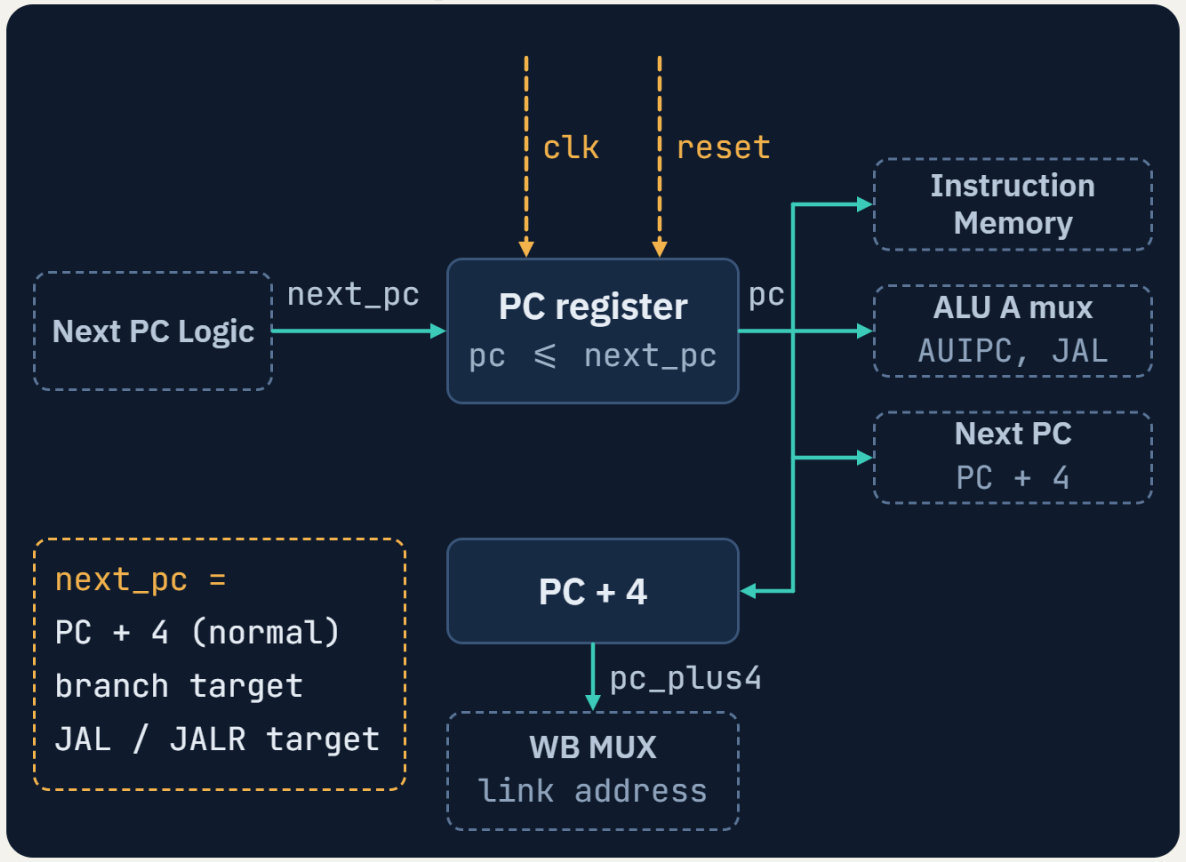
Format	Instructions	Opcode	Count
R-type	ADD, SUB, SLL, SLT, SLTU, XOR, SRL, SRA, OR, AND	0110011	10
I-type ALU	ADDI, SLTI, SLTIU, XORI, ORI, ANDI, SLLI, SRLI, SRAI	0010011	9
Load	LB, LH, LW, LBU, LHU	0000011	5
Store	SB, SH, SW	0100011	3
Branch	BEQ, BNE, BLT, BGE, BLTU, BGEU	1100011	6
Jump	JAL	1101111	1
Jump register	JALR	1100111	1
Upper immediate	LUI, AUIPC	0110111, 0010111	2
Total			37

FENCE, ECALL and EBREAK are not implemented.

FETCH

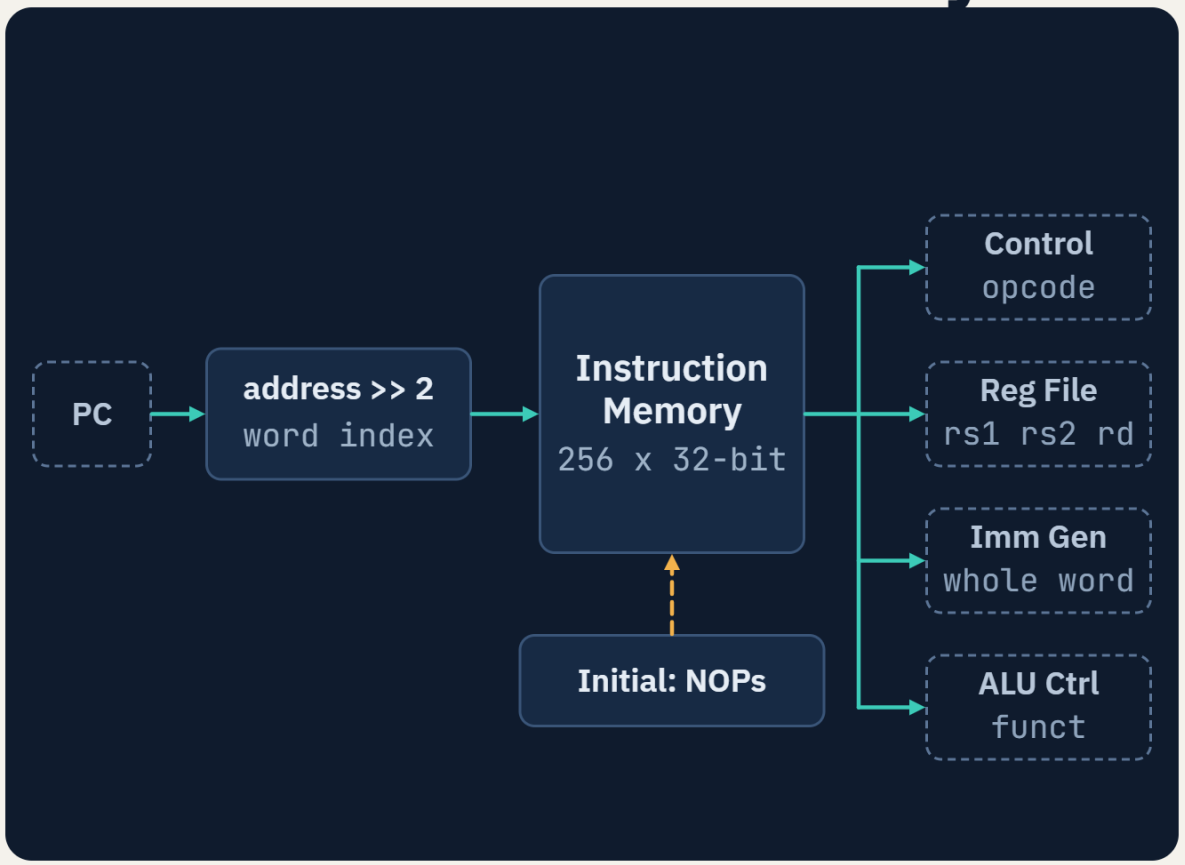
data ---- control other module

Program Counter



The PC loads next_pc on every clock edge and resets to 0. It is not always PC + 4: Next PC Logic picks the value, and pc_plus4 only supplies the JAL/JALR return address.

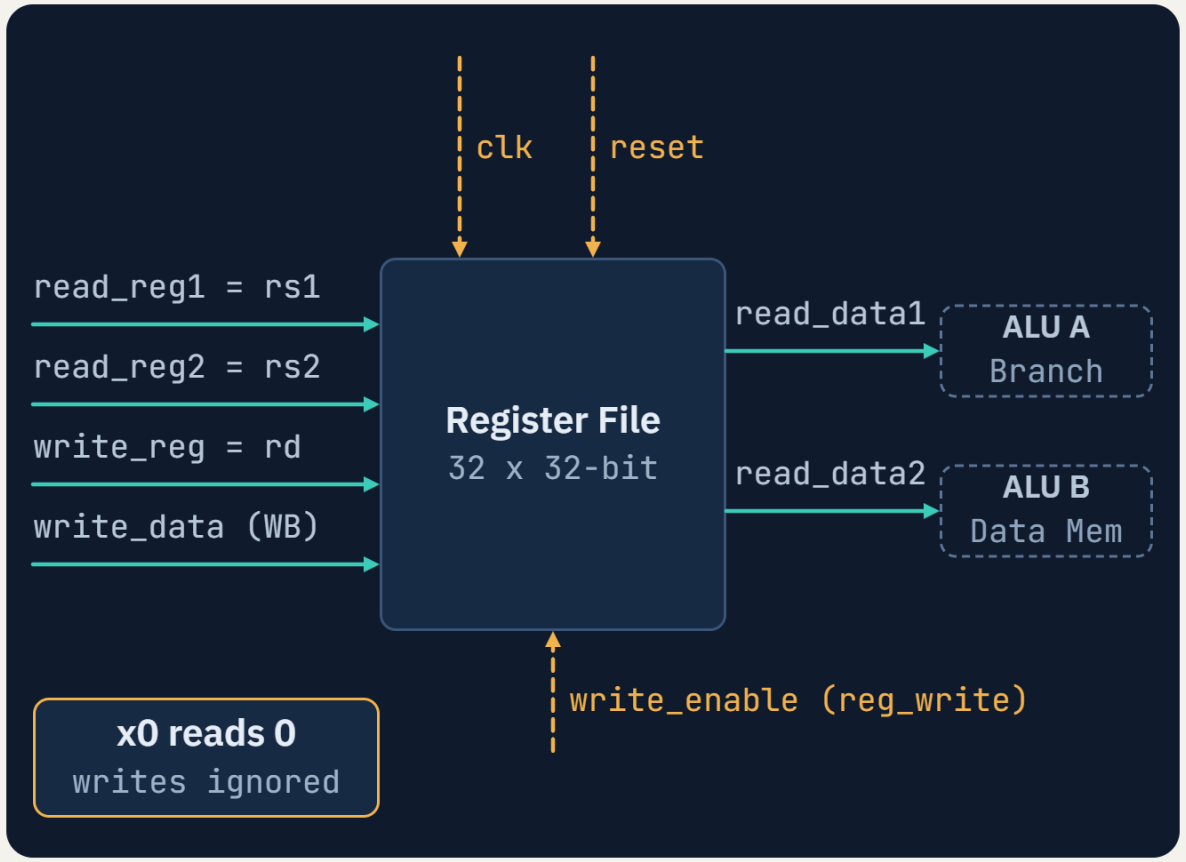
Instruction Memory



A 256-word array read combinationally; the byte address is shifted right by 2 to pick a word. The same instruction fans out to the Control Unit, registers, immediate generator and ALU Control.

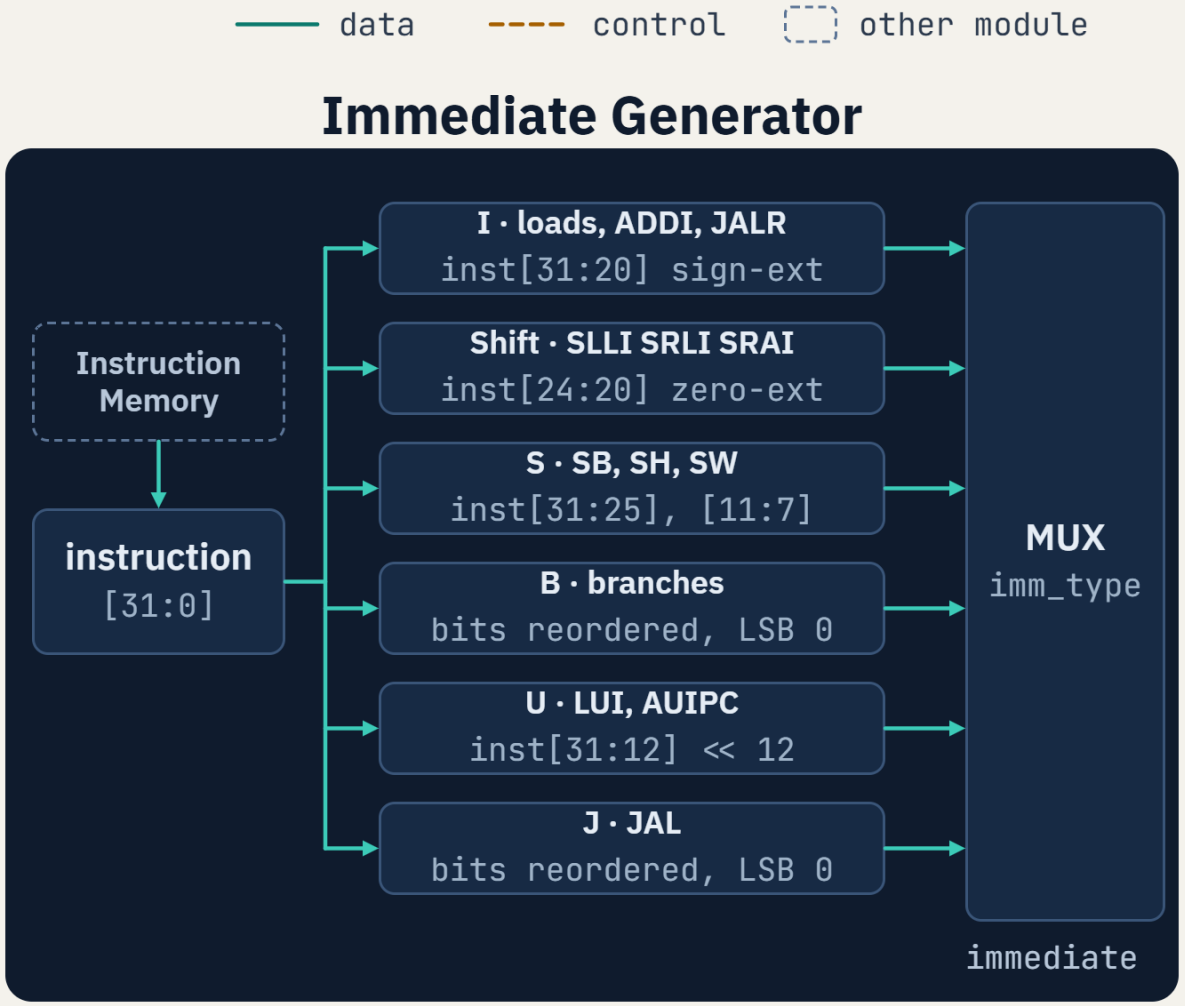
DECODE

Register File



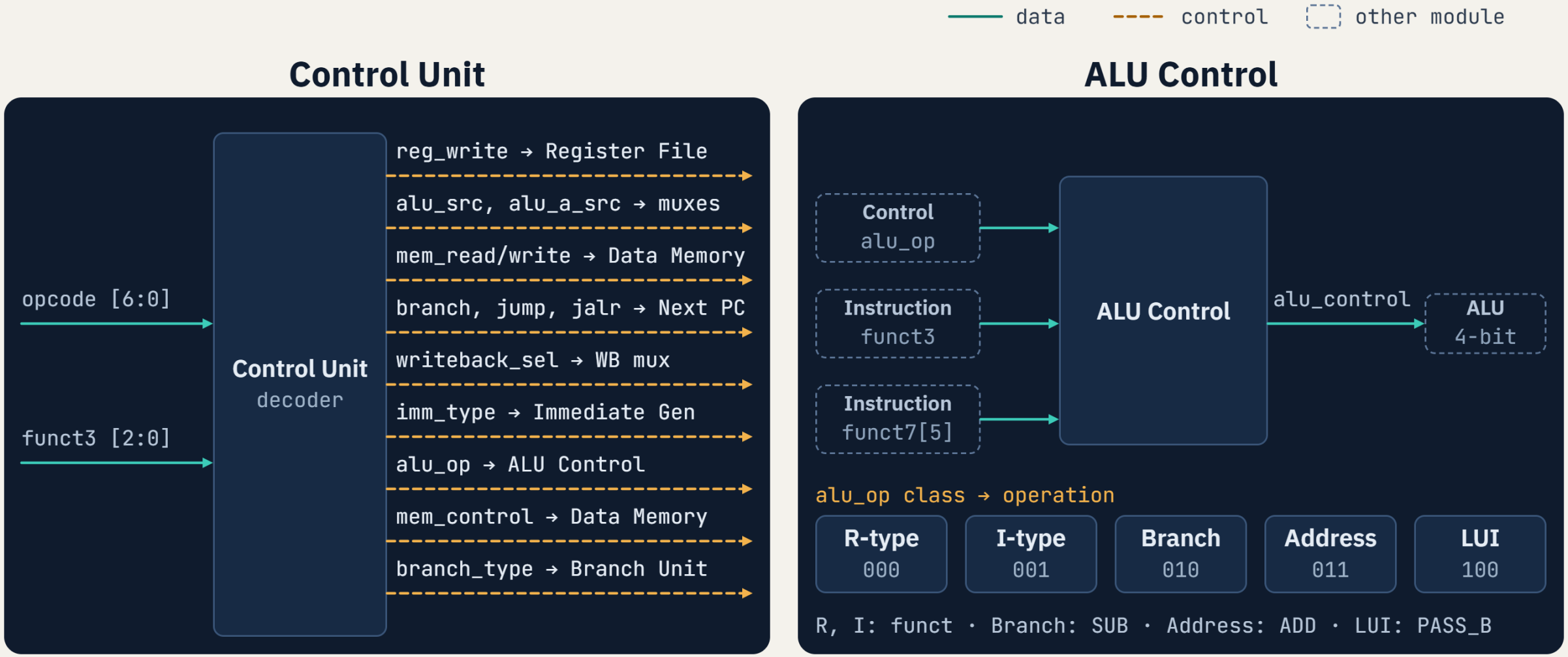
32 registers with two combinational read ports and one write port, clocked on the rising edge when reg_write is high. x0 always reads 0 and ignores writes.

Immediate Generator



Rebuilds the 32-bit immediate for the I, shift, S, B, U and J formats from scattered instruction bits, sign-extending from bit 31. B and J offsets get a 0 in bit 0.

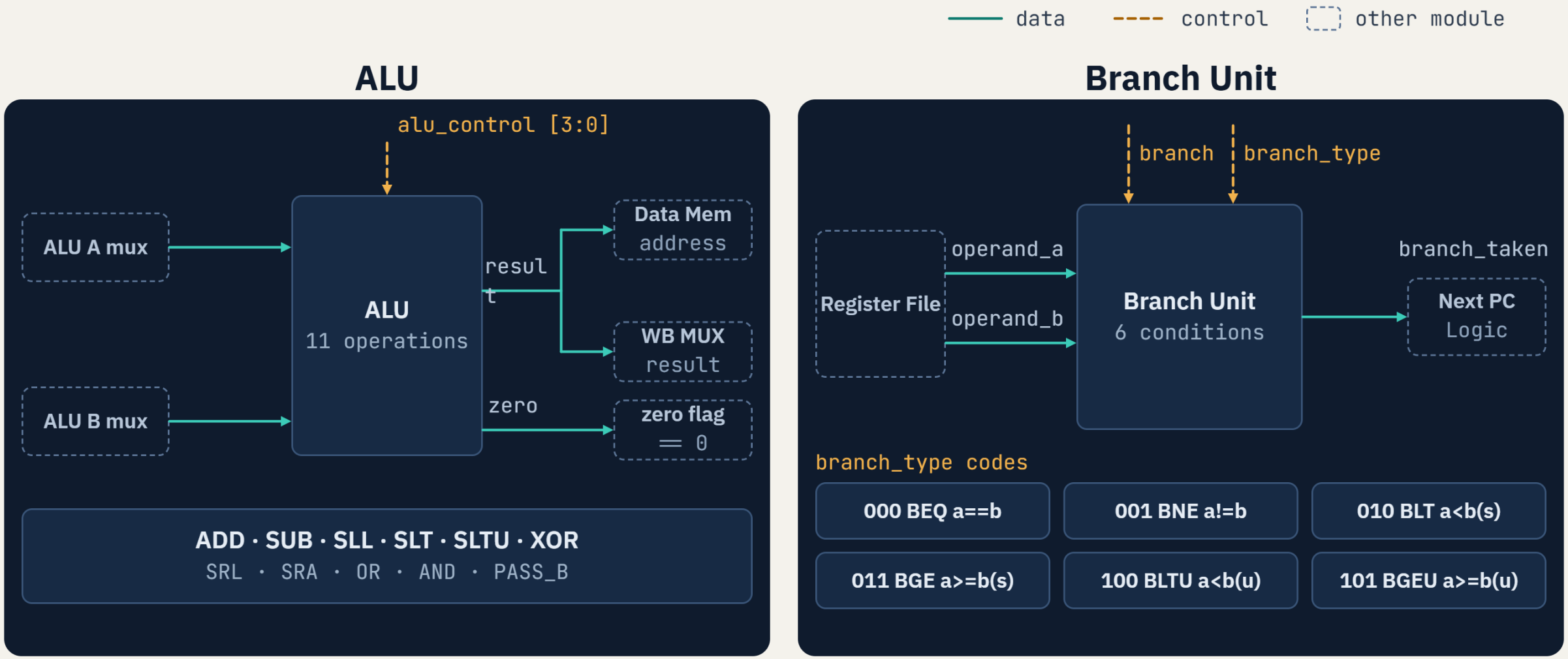
CONTROL



A combinational decoder: opcode (and funct3) in, every datapath control out. Unknown opcodes leave all controls at their safe default of 0, so nothing is written.

Turns the coarse alu_op class plus funct3 and funct7[5] into one of 11 ALU codes. funct7[5] separates ADD/SUB and SRL/SRA; branches use SUB and LUI uses PASS_B.

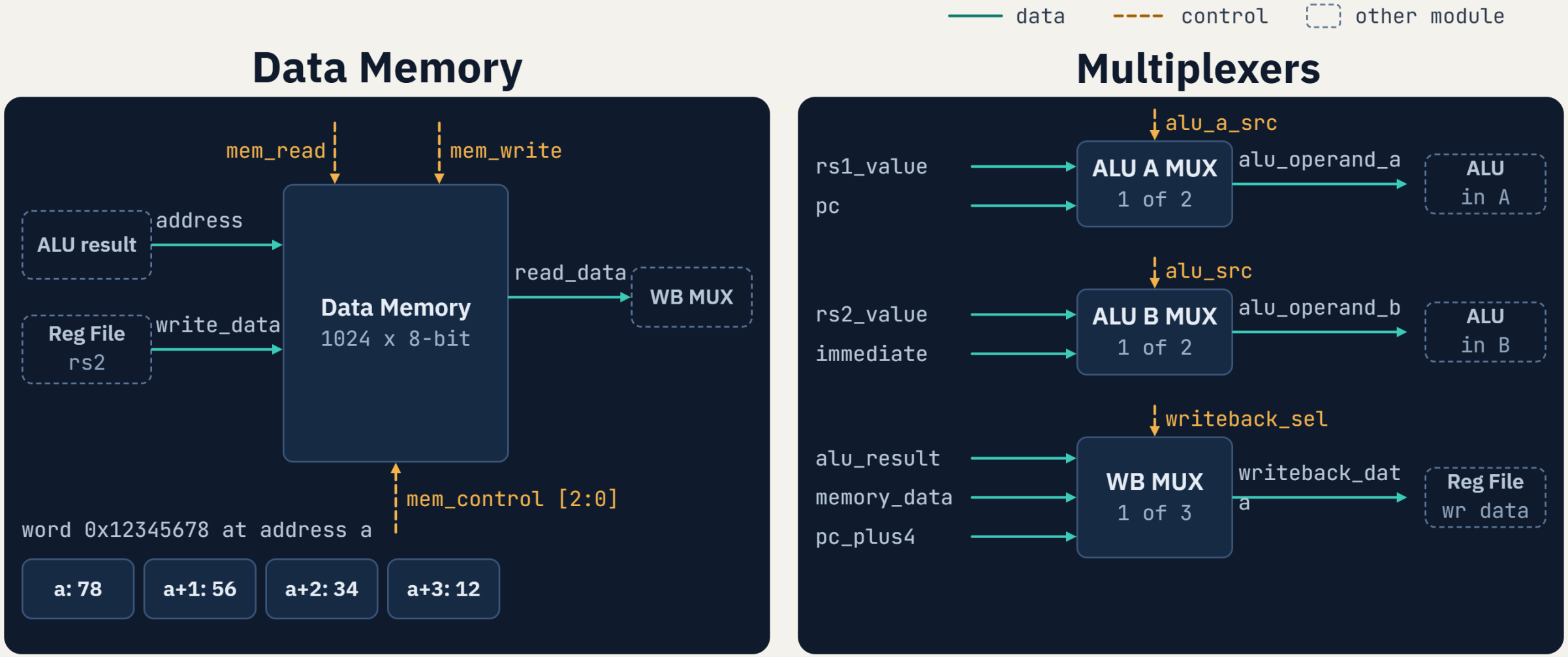
EXECUTE



Performs one of 11 operations on operands A and B. SLT is signed, SLTU unsigned, shifts use the low 5 bits of B, and PASS_B lets LUI load its immediate.

Compares rs1 and rs2 directly and raises branch_taken only when the branch control is set. BEQ and BNE test equality; BLT and BGE are signed; BLTU and BGEU are unsigned.

MEMORY AND DATA STEERING

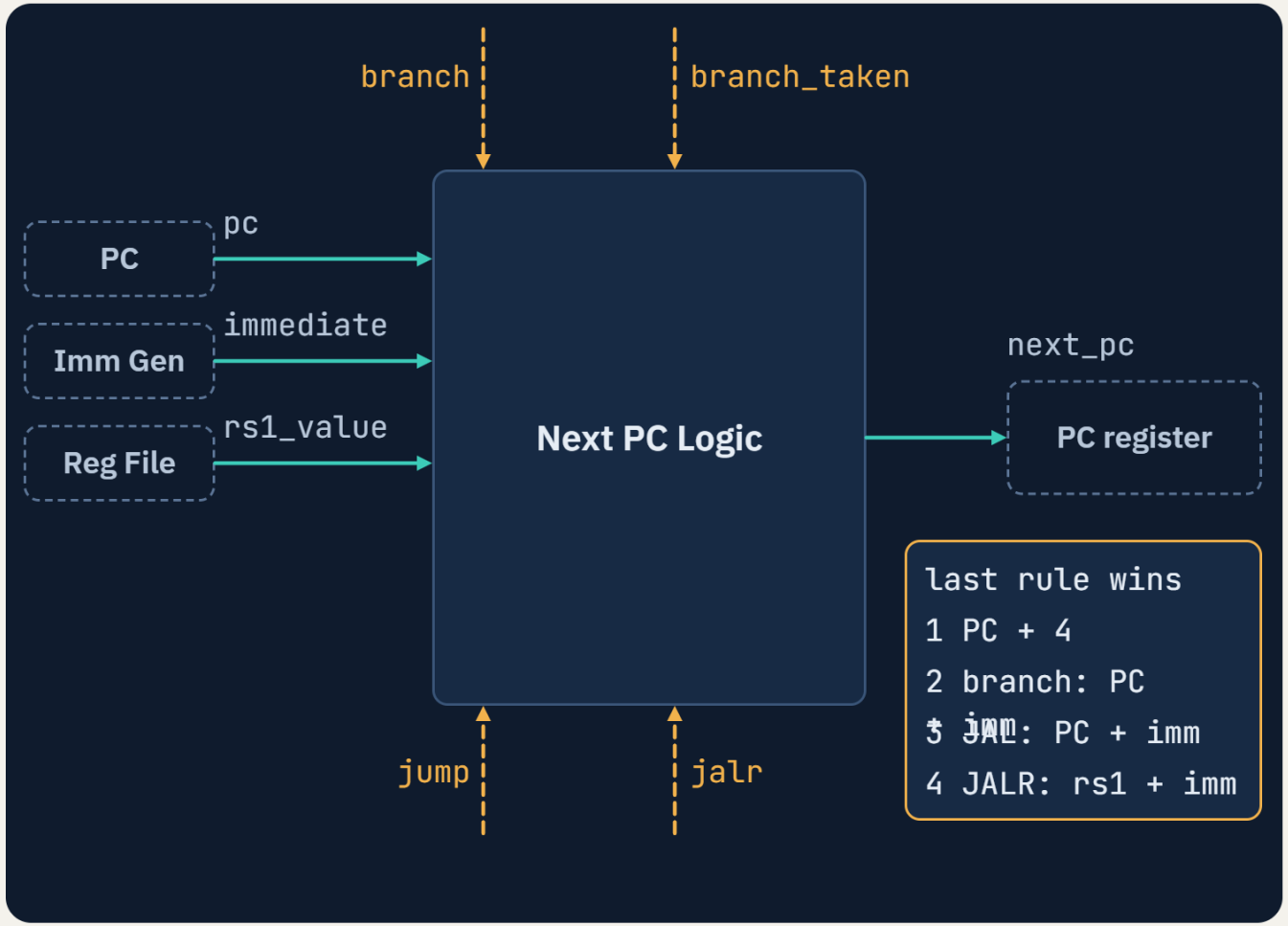


1024 bytes, little-endian. Stores write 1, 2 or 4 bytes on the clock edge; loads read combinationally, with LB/LH sign-extended and LBU/LHU zero-extended.

Three muxes steer data: ALU A picks rs1 or PC, ALU B picks rs2 or the immediate, and the WB mux picks ALU result, memory data or PC + 4 for the register write.

MODULE 11 · NEXT PC

Next PC Logic



data control other module

Chooses the address of the next instruction and feeds it back to the PC register. It starts with PC + 4 and then lets each control-transfer rule override the previous one, so the last matching rule wins.

KEY POINTS

- Taken branch: PC plus the B-type immediate
- JAL: PC plus the J-type immediate
- JALR: rs1 plus the I-type immediate, with bit 0 cleared

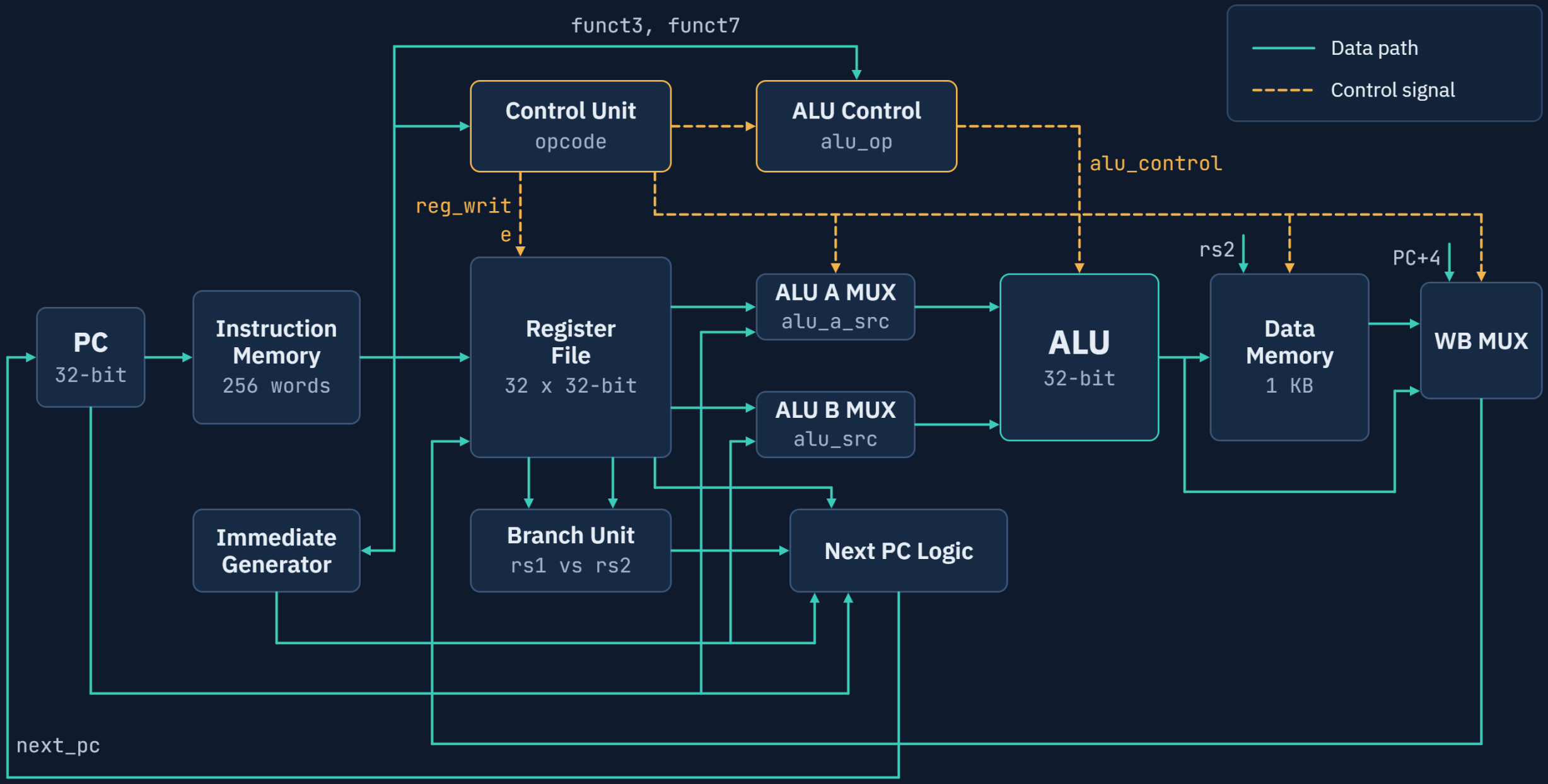
PUTTING IT TOGETHER

How Different Instructions Use the Modules

Instruction	ALU A	ALU B	ALU does	Data memory	Writes back	Next PC
ADD x3, x1, x2	rs1	rs2	add	idle	ALU result to x3	PC + 4
LW x24, 4(x21)	rs1	imm	add (address)	read word	memory to x24	PC + 4
SW x1, 4(x21)	rs1	imm	add (address)	write rs2	nothing	PC + 4
BEQ x1, x1, +8	rs1	rs2	SUB	idle	nothing	PC + imm if equal
JAL x28, +8	PC	imm	add	idle	PC + 4 to x28	PC + imm
JALR x28, x30, 0	rs1	imm	add	idle	PC + 4 to x28	(rs1 + imm) & ~1
LUI x27, 0x12345	rs1	imm	pass B	idle	ALU result to x27	PC + 4

Every instruction takes exactly one clock cycle.

Single-Cycle Datapath



SUMMARY

What the Design Achieves

What was built

- A modular RV32I CPU made of 12 Verilog modules
- 37 instructions across R, I, S, B, U and J formats
- A single-cycle datapath with one clock cycle per instruction
- A control unit plus a separate ALU control for clean decoding

Key design choices

- x0 is hard-wired to zero in the register file
- Branches are resolved by a dedicated Branch Unit
- Little-endian byte memory with sign and zero extension
- PASS_B in the ALU lets LUI reuse the same datapath

Thank you. Questions?